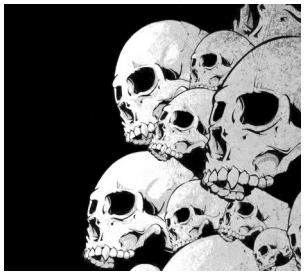


Guitarix



Y. Collette (ycollette.nospam@free.fr)
https://audinux.github.io

CC BY

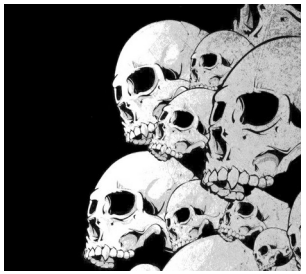


Guitarix

https://guitarix.org/



CC BY

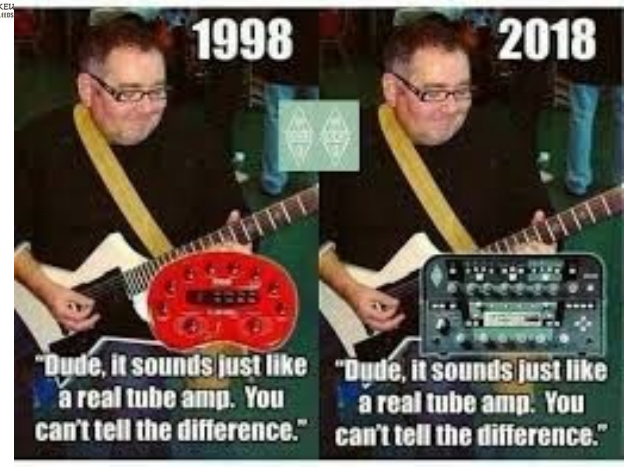


Some old ads



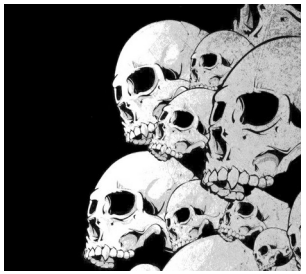








CC BY



Some old promises

In the 1980s, the "death of tubes" was heralded with transistor distortions, the Rockman, and later the SansAmp—a promised revolution, but tube amps never disappeared.

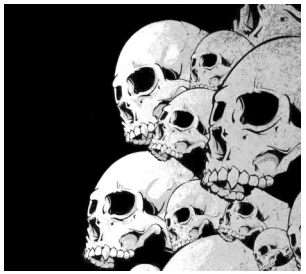
In the 2000s and 2010s, Line 6 PODs, then Kemper/Axe-FX, were heralded as "the death of the amp and pedalboard"—again, they mostly added a new palette of tools.

Today, with NAM, AIDA-X, Neural DSP, etc., the narrative is back: "plugins/AI replace everything"... but amps, pedals, and open-source solutions like Guitarix + IR + LADSPA/LV2 under Linux continue to offer excellent sounds.

“For 40 years, every new technology has promised to kill tubes and pedals… Yet, the good sounds were already there. What’s really changing is the accessibility and diversity of tools.”

“No need to wait for the next revolution: with Guitarix, IR and free plugins under Linux, we already have the tools to sculpt a professional sound.”

CC BY



Guitarix
General appearance

Activation of different windows

Effects available

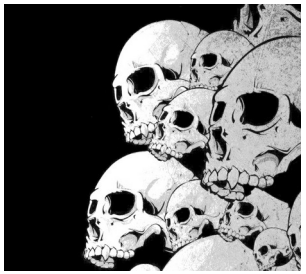
General parameters

Tuner

Effect rack

Presets





Guitarix
The buttons bar

Pool: activates / deactivate the display of the effect list bar.

Order: Reduces the height of all effect modules and allows you to easily click and reorganize these effects. This view is practical to reorganize large amounts of effects.

Tuner: Activates / deactivate the tacking of the tuning rack.

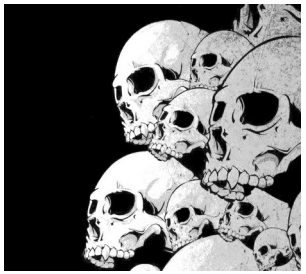
Rack: Activates / deactivate the Display of Effect Modules Rack.

Split: displays the modules rack in two columns, mono grafts on the left column and stereo grafts on the right column.

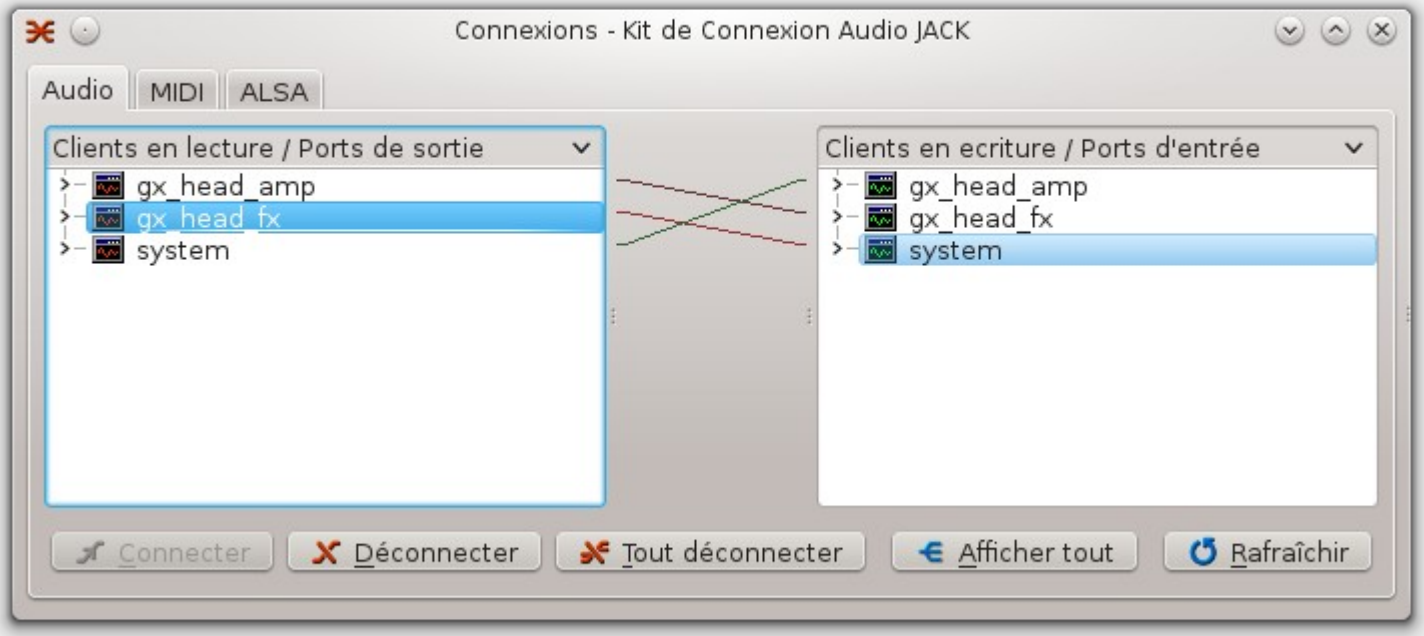
Fold / Show: the button less reduces the height of all effect modules. The button plus dilates.

Live: Displays a minimum full screen poster with preset information. Practical for live performance.

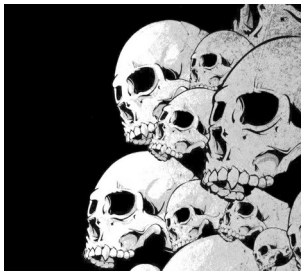
CC BY



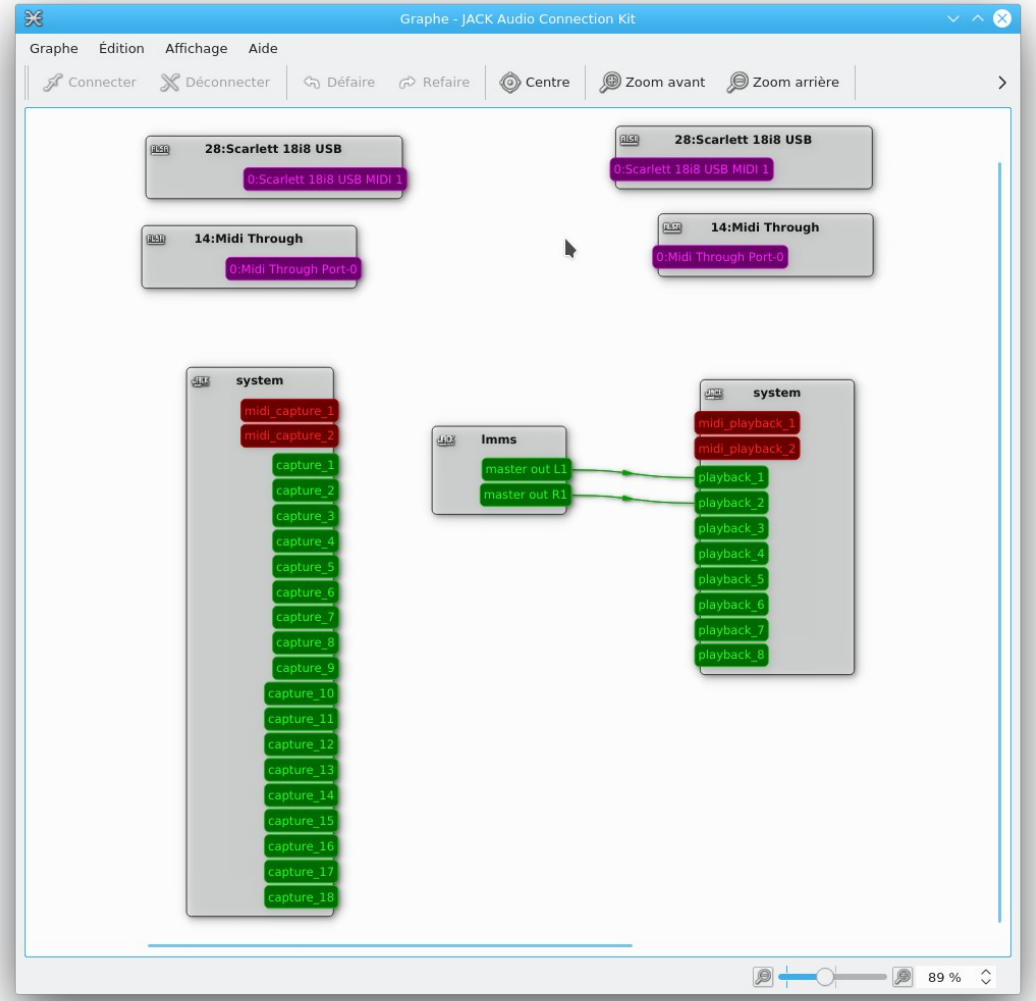
Guitarix
Jack side



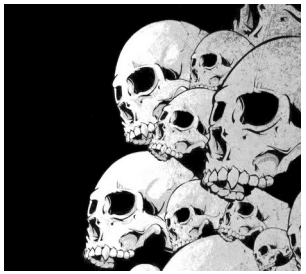
CC BY



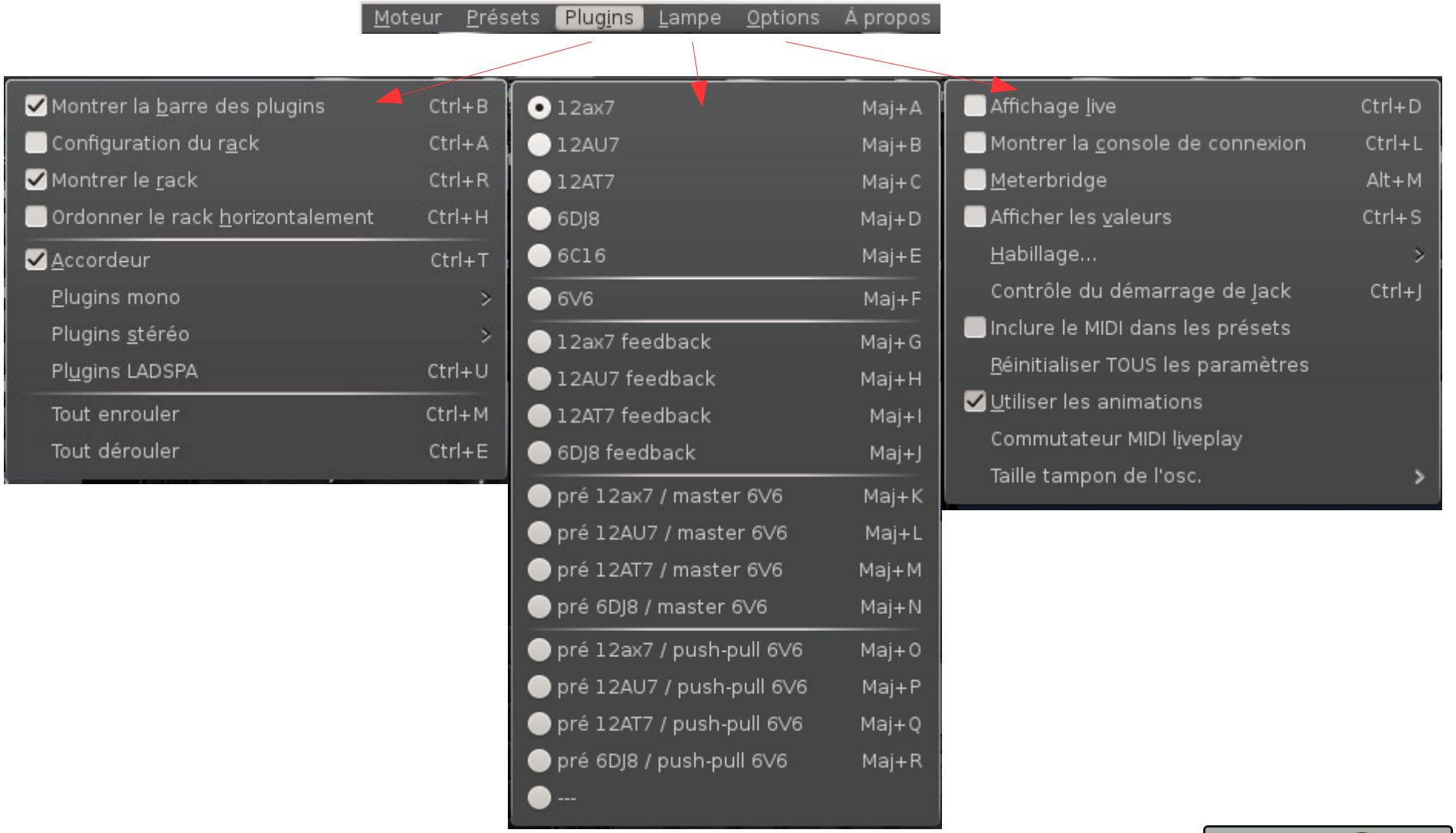
Guitarix
Jack side - New connection window



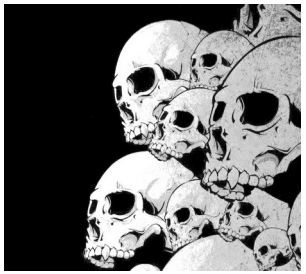
CC BY



Guitarix
The Menus



CC BY



Guitarix
The structure of a rack

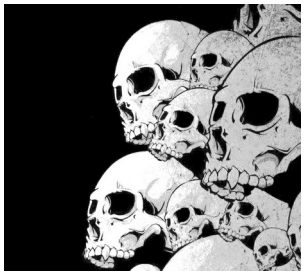
Pre-amplifier effects

Amplifier

Post-amplifier effects



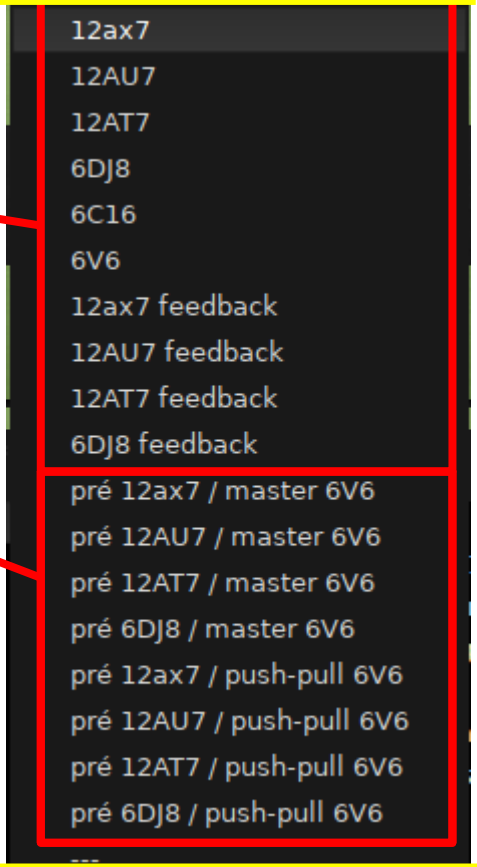
CC BY



The tube amp emulator



Setting with lamp choice just for the amplifier



General volume

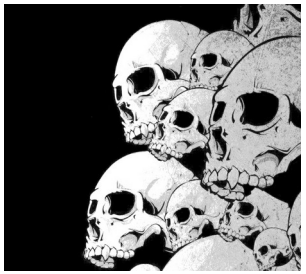
Sound characteristics settings

Mixture between signal with and without distortion

Signal gain

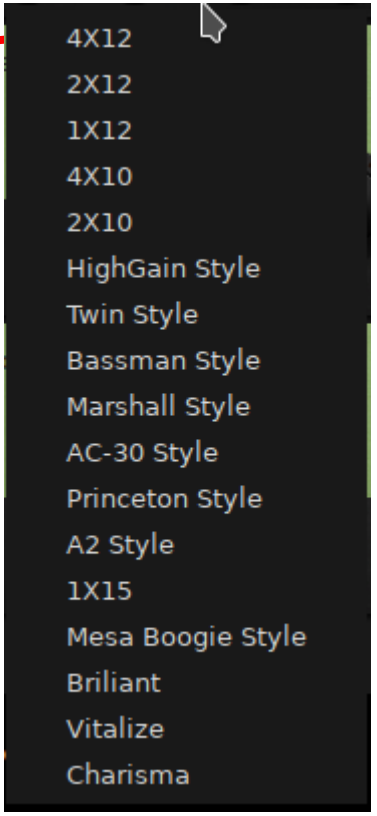
Setting with choice of lamp for pre-amplifier / amplifier

CC BY



Cabinet





A cabinet is a baffle simulator.
It allows you to simulate different types of amplifier or a speaker combination.

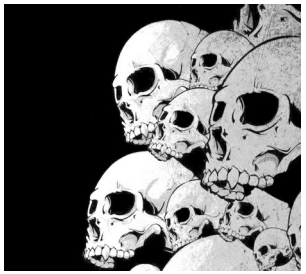


Cabinet 4x12



Cabinet Princeton style

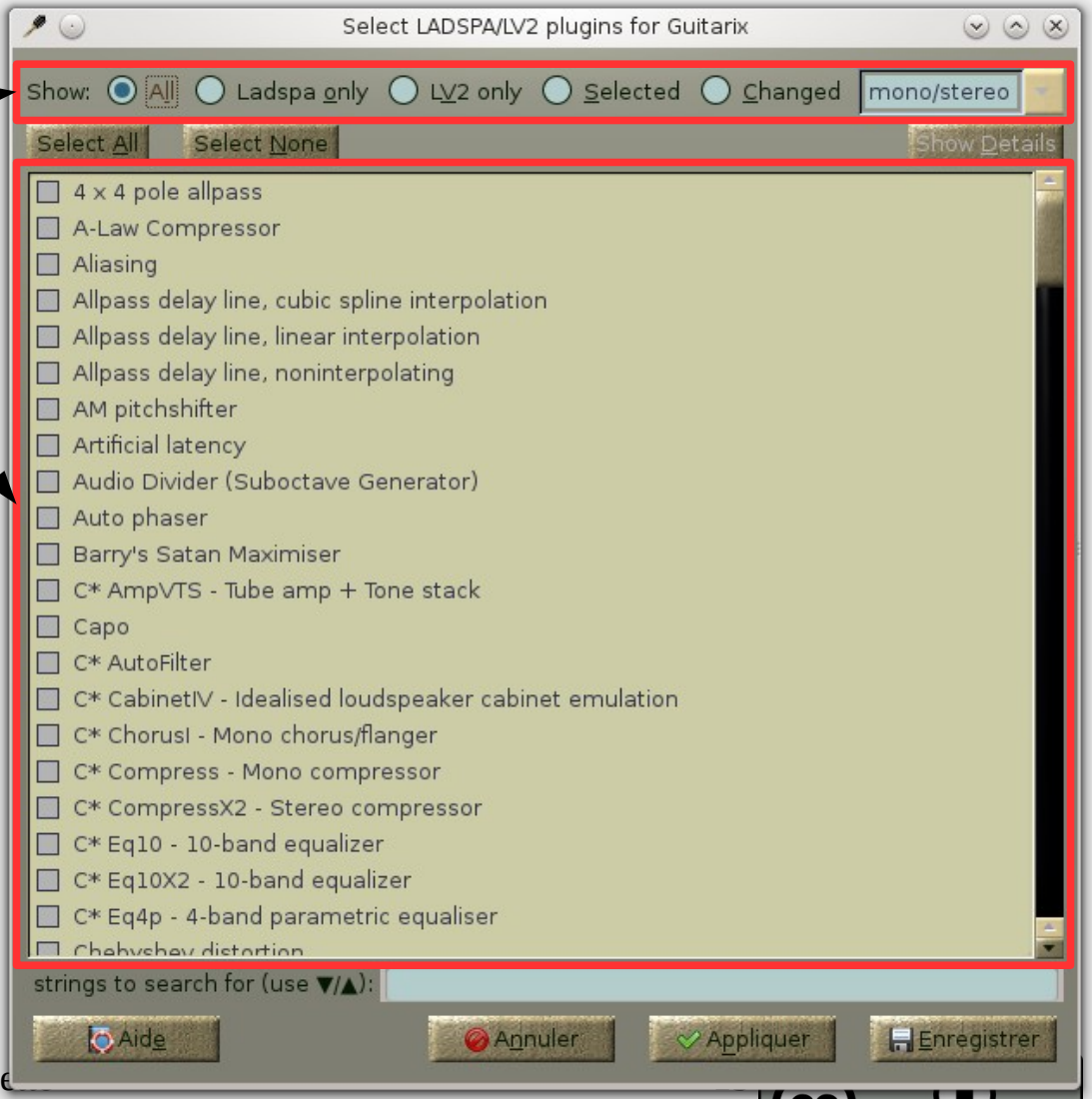
CC BY



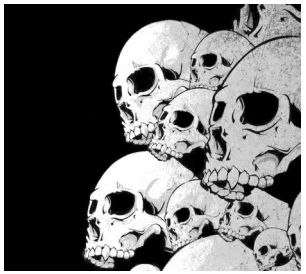
Plugin management LADSPA / LV2

Plugin type selection area (LADSPA / LV2)

List and selection of plugins to add to rack



CC BY



Plugins for guitar

Interesting plugins for the guitar:

https://portalmod.com

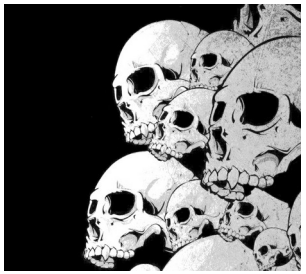
A distortion in LV2 format from Portalmod:

```
$ git clone https://github.com/portalmod/mod-distortion.git
$ cd mod-distortion
$ make # compilation
$ make install # Installation in the user directory
$ sudo dnf install mod-distortion
```

Tonality pedals in LV2 format from Portalmode:

```
$ git clone https://github.com/portalmod/mod-pitchshifter.git
$ cd mod-pitchschifter
$ make # compilation
$ make install # Installation in the user directory
$ sudo dnf install mod-gxpitchschifter
```


CC BY



Check via MIDI





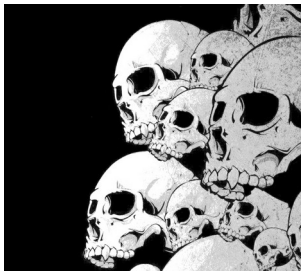
Objective: control an effect button via a MIDI instrument



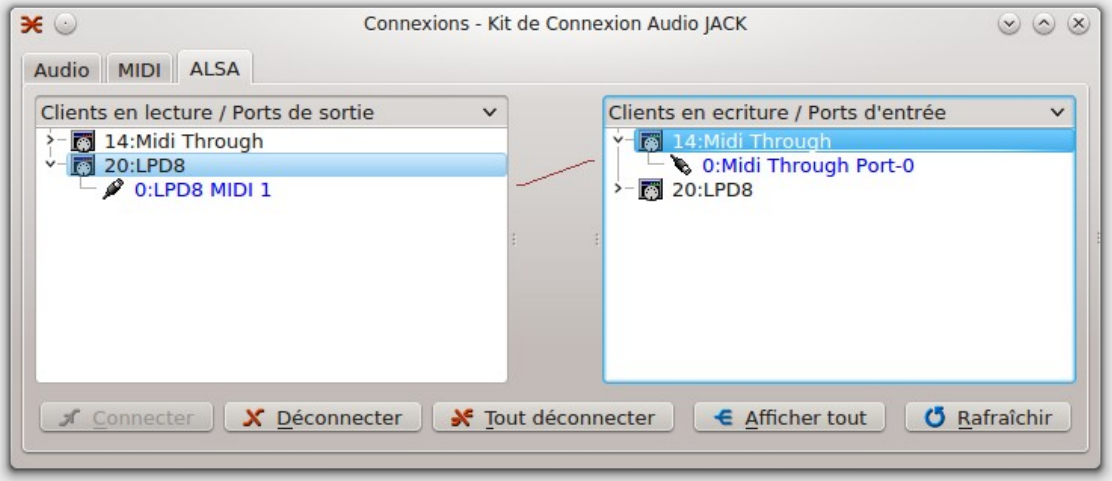




CC BY

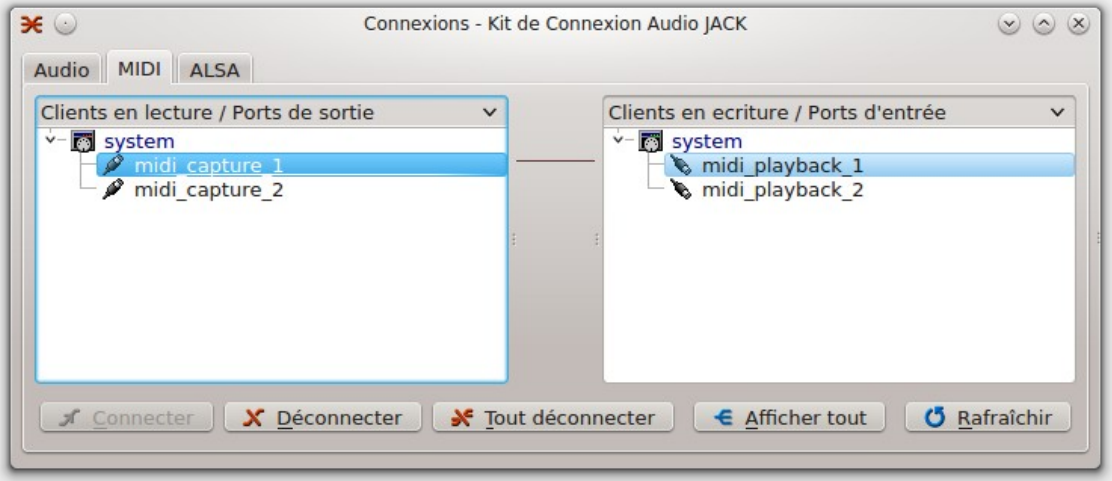


Control via Midi Jack connections



First step:
Alsa tab:

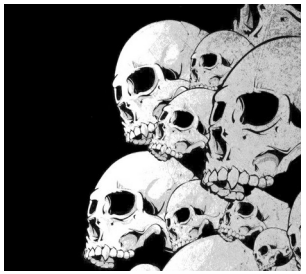
We connect the MIDI output of the LDP8 to the System system input: MIDI_PLAYBACK_1



Second step:
Midi tab:

We connect the system system output: MIDI_CAPTE_1 to the system system output: MIDI_PLAYBACK_1

CC BY



Control via Midi
MIDI / Guitarix association

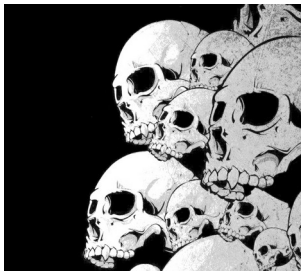


A click on the guitarix button to control with the middle of the mouse button or on the two buttons for a two buttons mouse reveals this dialog window.
Then you have to move the MIDI button you want to use. The references of this MIDI button are then recovered by guitarix.
You have to click on "Validate" to record the Association Midi / Guitarix button.



The engine menu → MIDI controller reveals this dialogue.
By clicking on "Including MIDI controllers in the Presets" allows you to save MIDI / Guitarix associations when you save the presets.

CC BY



The Dubber

Reading sense

Overdub

Mix between signal and dubber

Registration



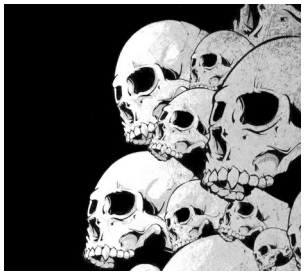
Start and end adjustment

Delete

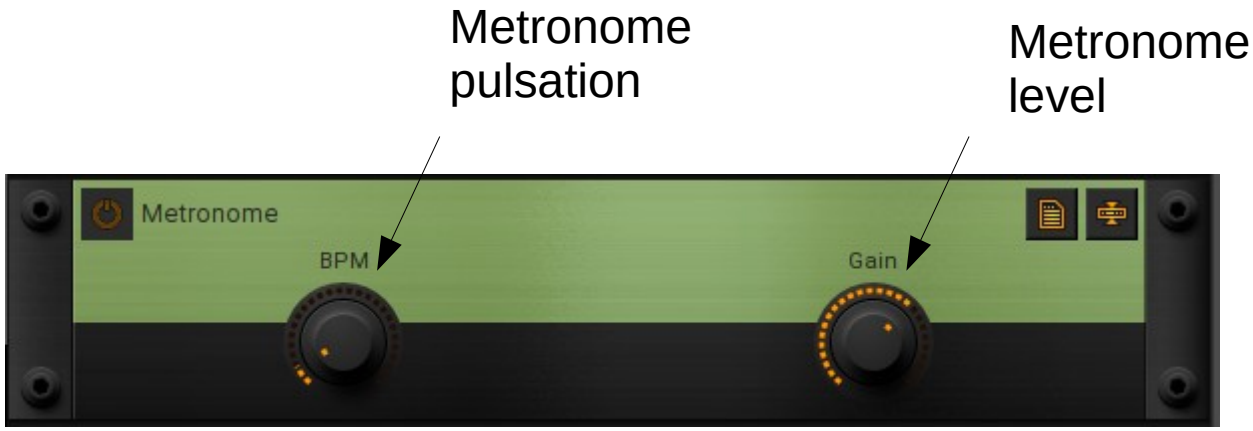
Dubber Level

The Dubber allows you to record a sound loop and to replay it

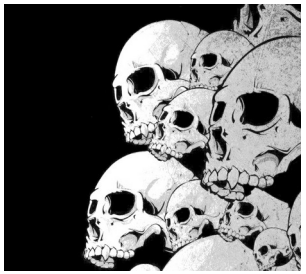
CC BY



The metronome



The metronome diffuses a beat. The sound level of the beat is proportional to the level of the sound at the input.



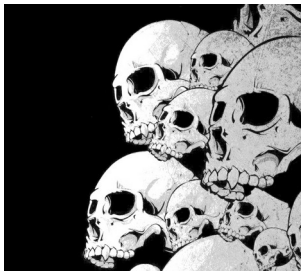
Guitarix and web server 1/2

Guitarix can be launched without graphical interface.
It is then controllable via a web server.
This web server can then be checked via a graphics tablet for example.

Guitarix compilation in web server mode:

```
# We recover the source code of guitarix (Git must be installed)
$ git clone https://github.com/brummer10/guitarix
# We initialize guitarix dependencies
$ git submodule update --init --recursive
$ cd trunk/webui/bootplate/
$ git submodule update -init --recursive
$ cd ../ ..
# We are now compiling guitarix (GCC must be installed)
$ ./waf configure -prefix =/opt/guitarix-Devel
$ ./waf
# We install everything
$ su -c ./waf install
```


CC BY



Guitarix and web server 2/2

1. Starting guitarix:

```
$ /opt/guitarix-devel/bin/guitarix -N
```

2. start the websockify program from the current directory (webui):

```
$ python -m websockify.websocketproxy --web=. '*':8000 localhost:7000
```

3. open the URL http://localhost:8000 in the desktop browser

You should be redirected to debug.html. index.html expects preprocessed files which can be generated with the deploy script:

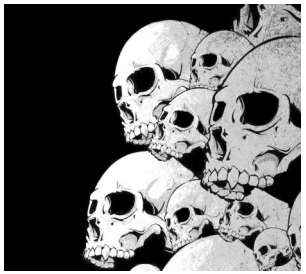
```
$ ./deploy.sh
```

This script creates the build/ directory which is used in index.html.
deploy.sh needs the node program from node.js. For Debian/Ubuntu:

```
$ su -c dnf install nodejs
```

Next step :
- open the port 8000 on the tablet and on the server ;
- connect to the guitarix web server via the tablet ;
- install this web server on the Raspberry Pi.

CC BY

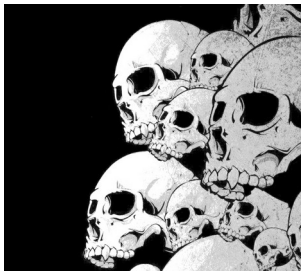


Guitarix Scope

The Scope allows you to visualize the signal to a given place in the rack.
It is practical because it makes it possible to visualize the influence of the settings of an effect.
In this way, we can dose the quantity of distortion.
Too much gain will generate an almost square signal.
Not enough gain, the signal will remain sinusoidal.
It remains to find the right dose.



CC BY

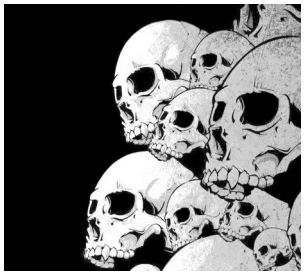


Guitarix / VST3

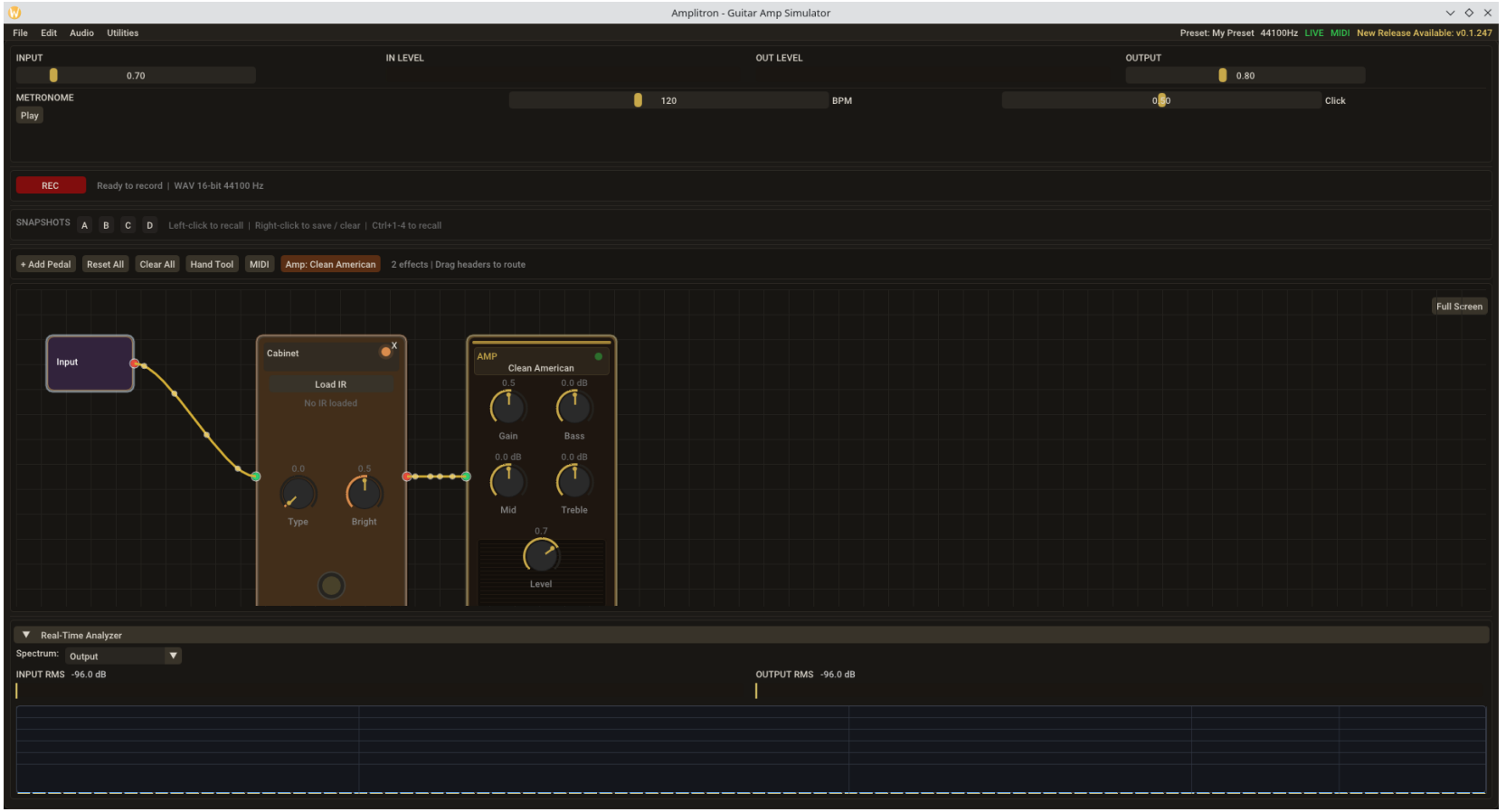
It’s now possible to use Guitarix as a VST3 plugin in Carla for example



CC BY



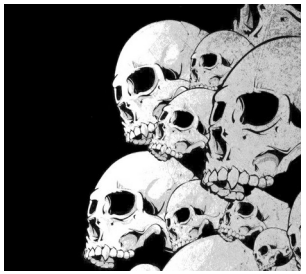
Amplitron



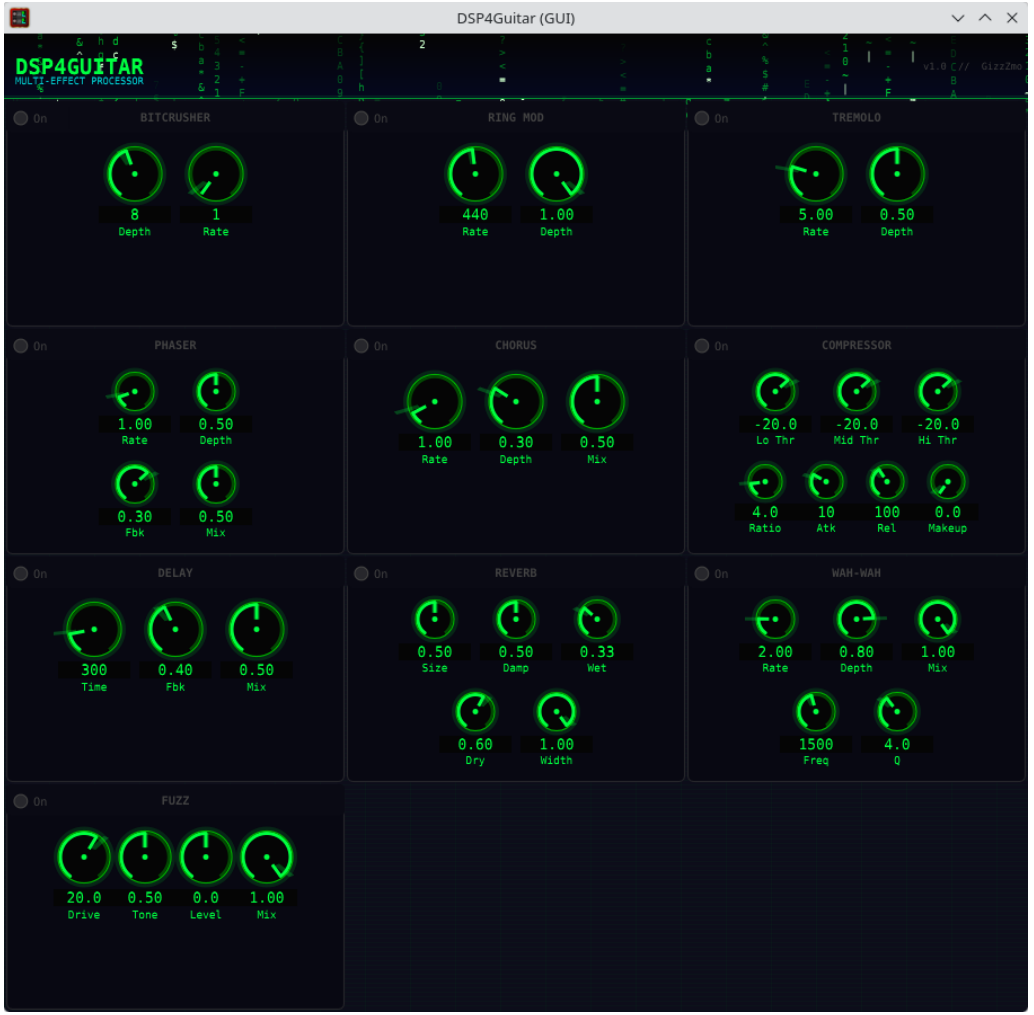
$ dnf install amplitron

https://github.com/sudip-mondal-2002/Amplitron

CC BY



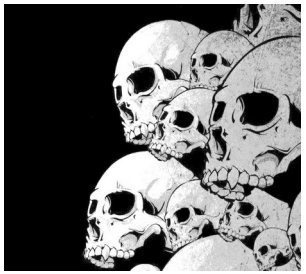
DSP4Guitar



$ dnf install vst3-dsp4guitar

https://github.com/GizzZmo/DSP4Guitar

CC BY



SpiceFX

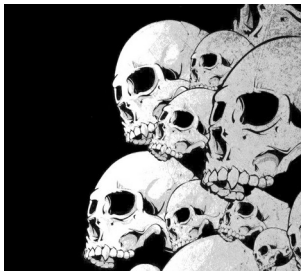


$ dnf install lv2-spicefx
$ dnf install vst3-spicefx

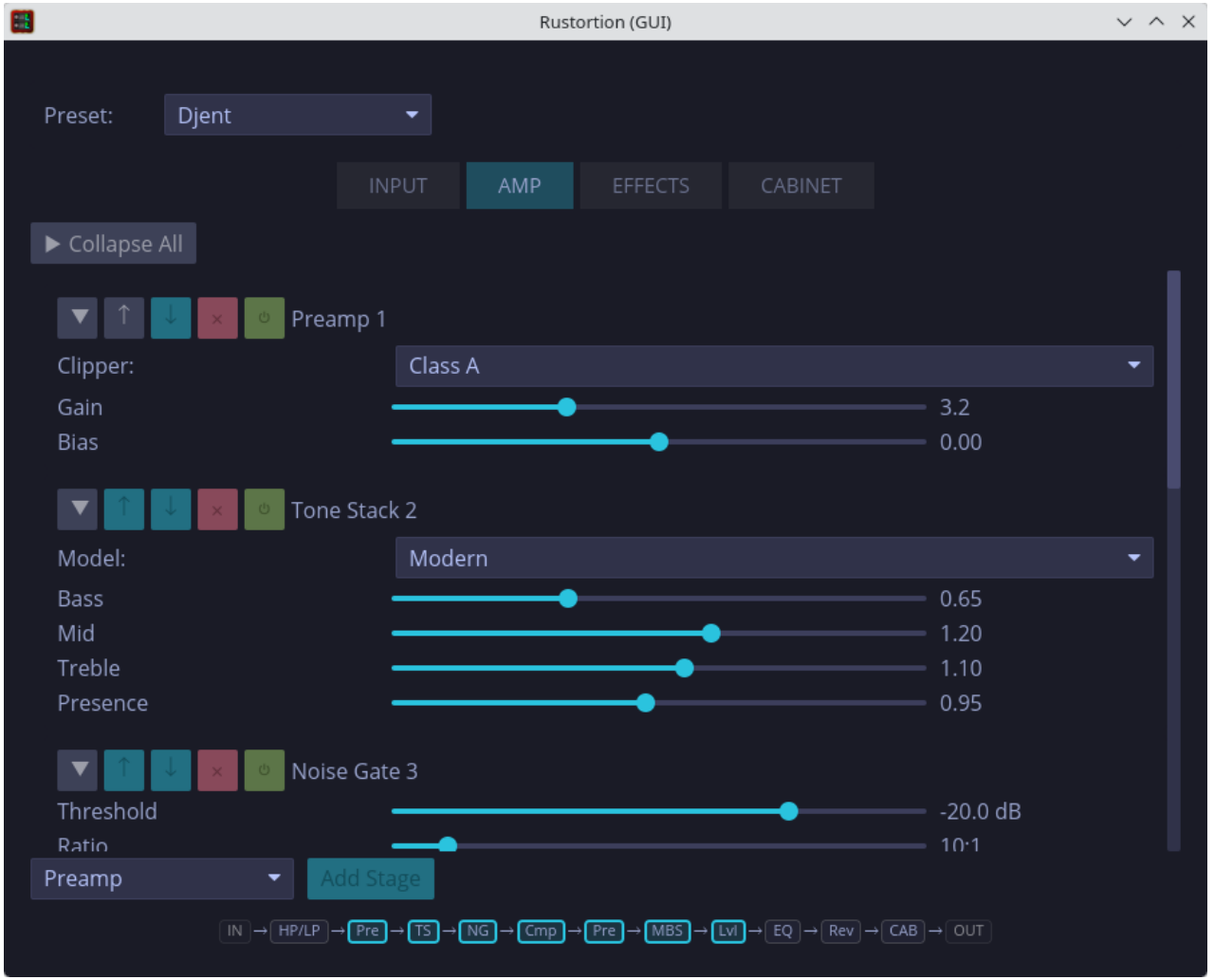
https://fr.wikipedia.org/wiki/Fou_litt%C3%A9raire

https://github.com/DatanoiseTV/spice-oss

CC BY



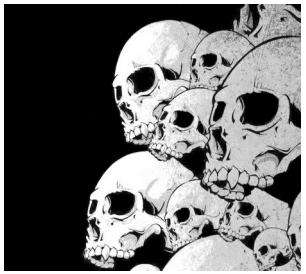
Rustortion



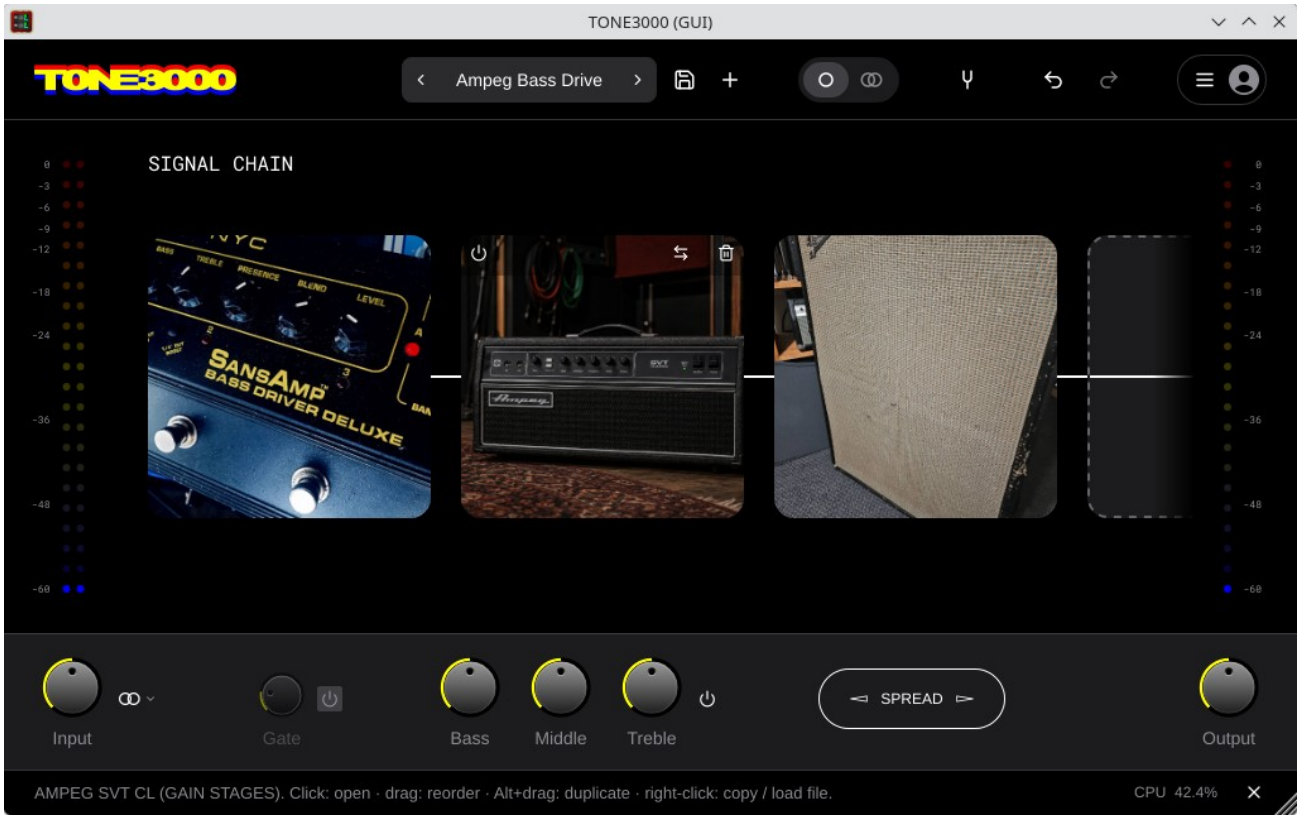

```
$ dnf install rustortion
$ dnf install clap-rustortion
$ dnf install vst3-rustortion
```

https://github.com/OpenSauce/rustortion

CC BY



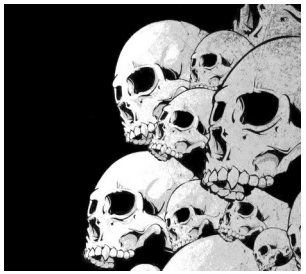
Tone3000




```
$ dnf install tone3000
$ dnf install clap-tone3000
$ dnf install vst3-tone3000
$ dnf install lv2-tone3000
```

https://github.com/tone-3000/tone3000-plugin

CC BY



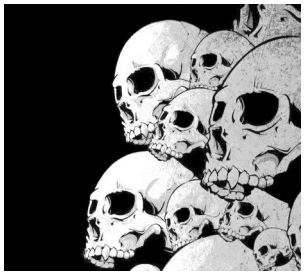
open_riff_box



https://github.com/dlujic/open-riff-box

```
$ dnf install open_riff_box
$ dnf install vst3-open_riff_box
```


CC BY



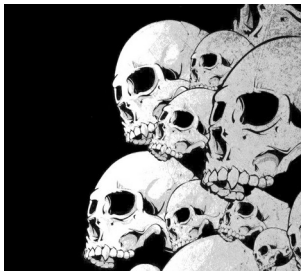
AmpForge




```
$ dnf install ampforge
$ dnf install vst3-ampforge
$ dnf install clap-ampforge
$ dnf install lv2-ampforge
```

https://github.com/Loursy/AmpForge

CC BY



Muscriptor Introduction

MuScriptor is a multi-instrument music transcription model developed by Kyutai and Mirelo. You will need a HuggingFace Token and a granted access to :
https://huggingface.co/MuScriptor/muscriptor-medium
https://huggingface.co/MuScriptor/muscriptor-large
https://huggingface.co/MuScriptor/muscriptor-small

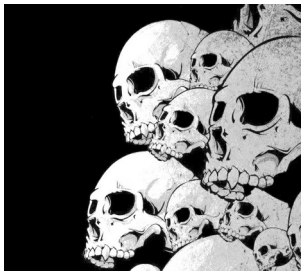
The github repository of Muscriptor :
https://github.com/muscriptor/muscriptor

Create a Python virtual environment :
$ python3 -m venv muscriptor

Install muscriptor via pip (uses 4.8 Go of disk space):
$ source muscriptor/bin/activate
$ pip install muscriptor

Launch muscriptor :
$ source muscriptor/bin/activate
$ muscriptor transcribe -f midi -m medium Song.mp3

CC BY



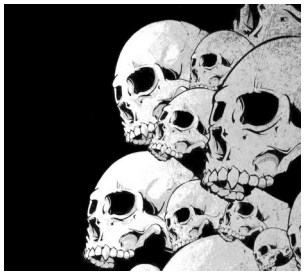
Webography

You can access to a huge set of presets for Guitarix on Musical Artifacts :

https://musical-artifacts.com/artifacts?apps=guitarix

All these artifacts can be easily downloaded into Guitarix because Musical Artifacts is now integrated into Guitarix.

CC BY



Documentary

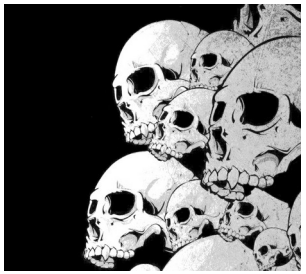


A good documentary about pedals, who made them …

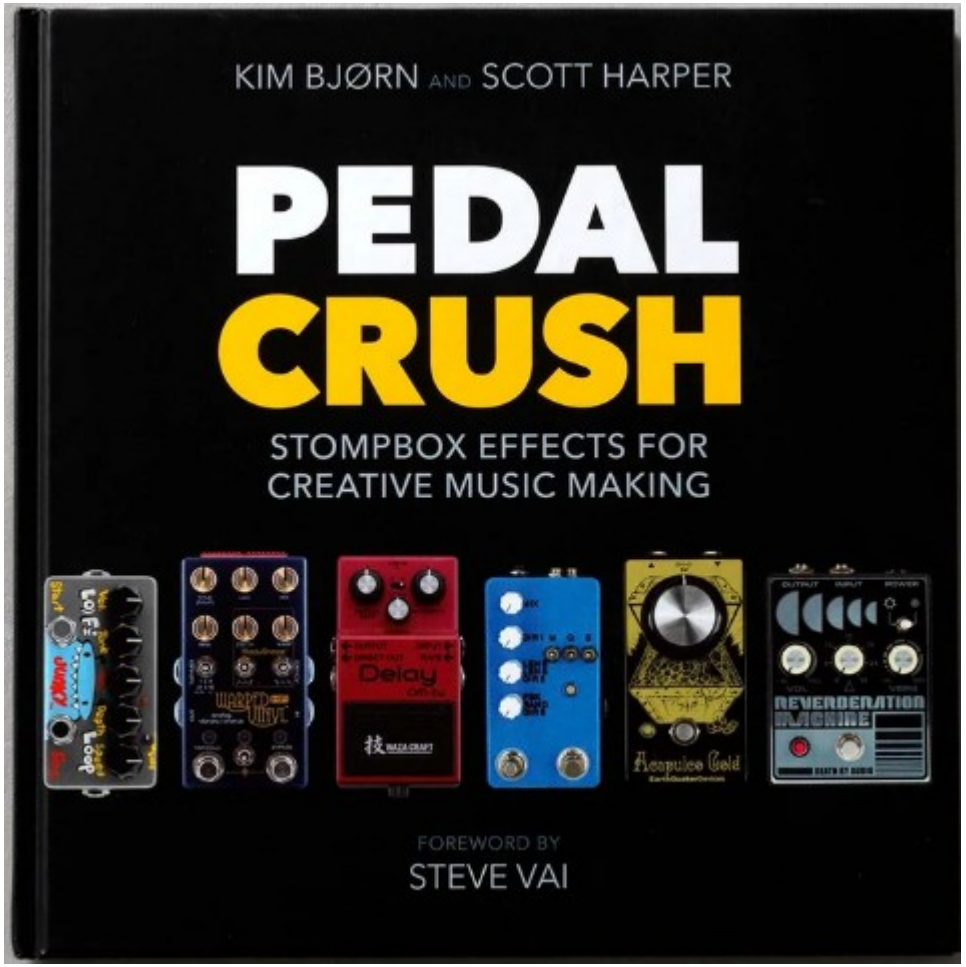
The PEDAL Movie :

https://www.imdb.com/fr/title/tt11811310/

CC BY



Book



A good book about pedals :

PEDAL CRUSH

https://bjooks.com/en-fr/products/pedal-crush-stompbox-effects-for-creative-music-making

CC BY